Supporting Information

Predicting Salt Effects on the Binding of PFAS to Protein

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Figures

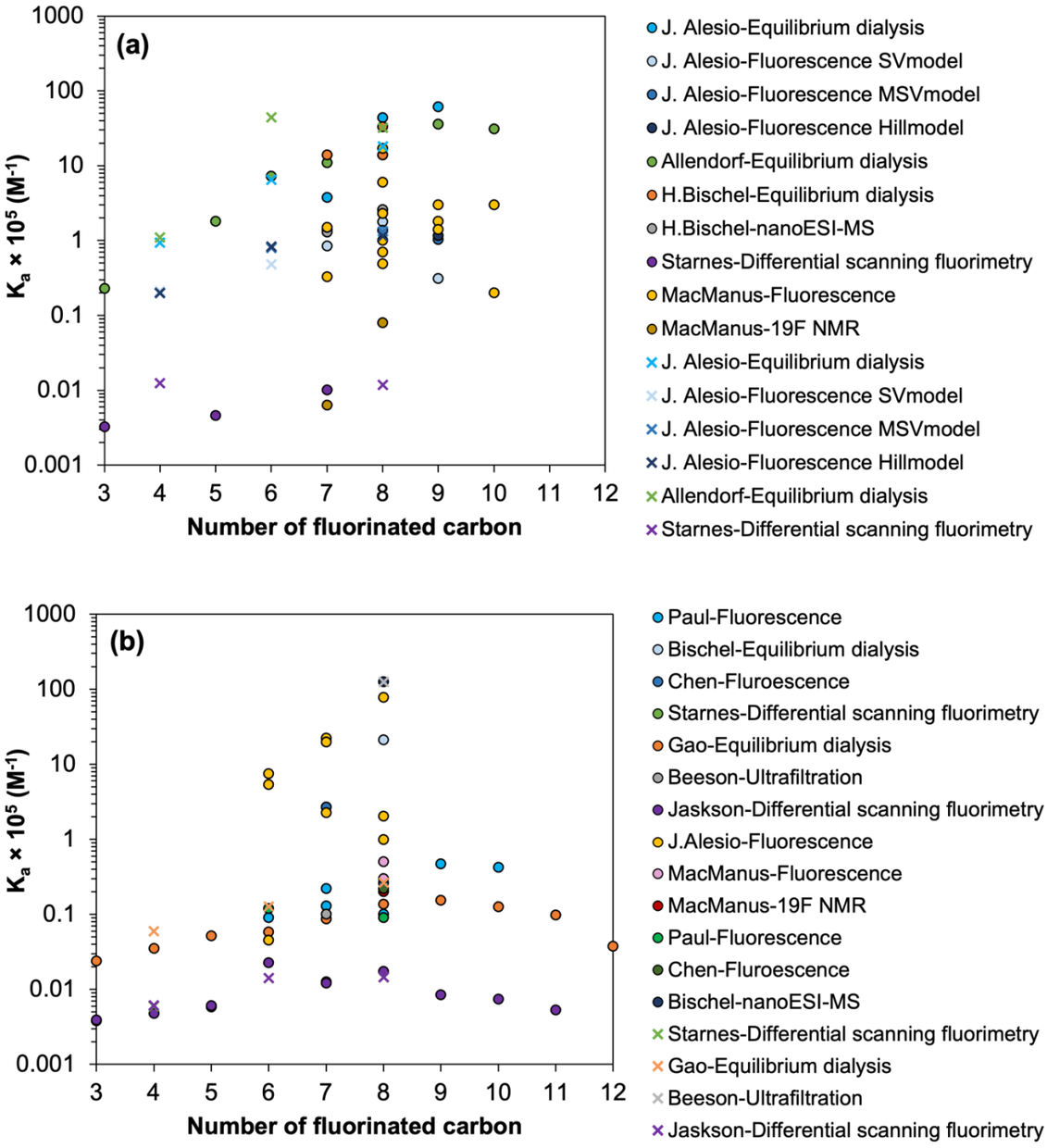


Figure S1. Comparison of binding constant, Ka, from literature studies for PFAS interactions with (a) BSA and (b) HSA[1-9]. Circle symbols represent PFCAs, while cross symbols represent PFSAs.

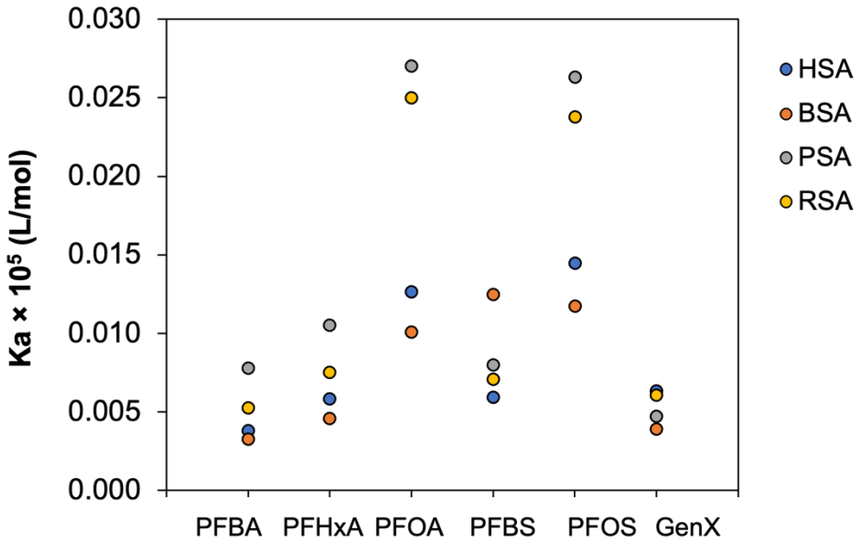


Figure S2. Comparison of binding constant, Ka, from Starnes, et al. [5]for PFAS interactions with various serum albumins.

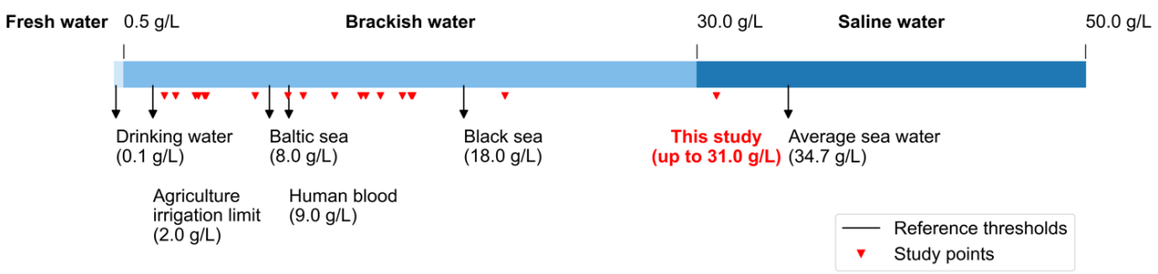


Figure S3. Water salinity scale in the environment compared to the range of salt concentration in this study.

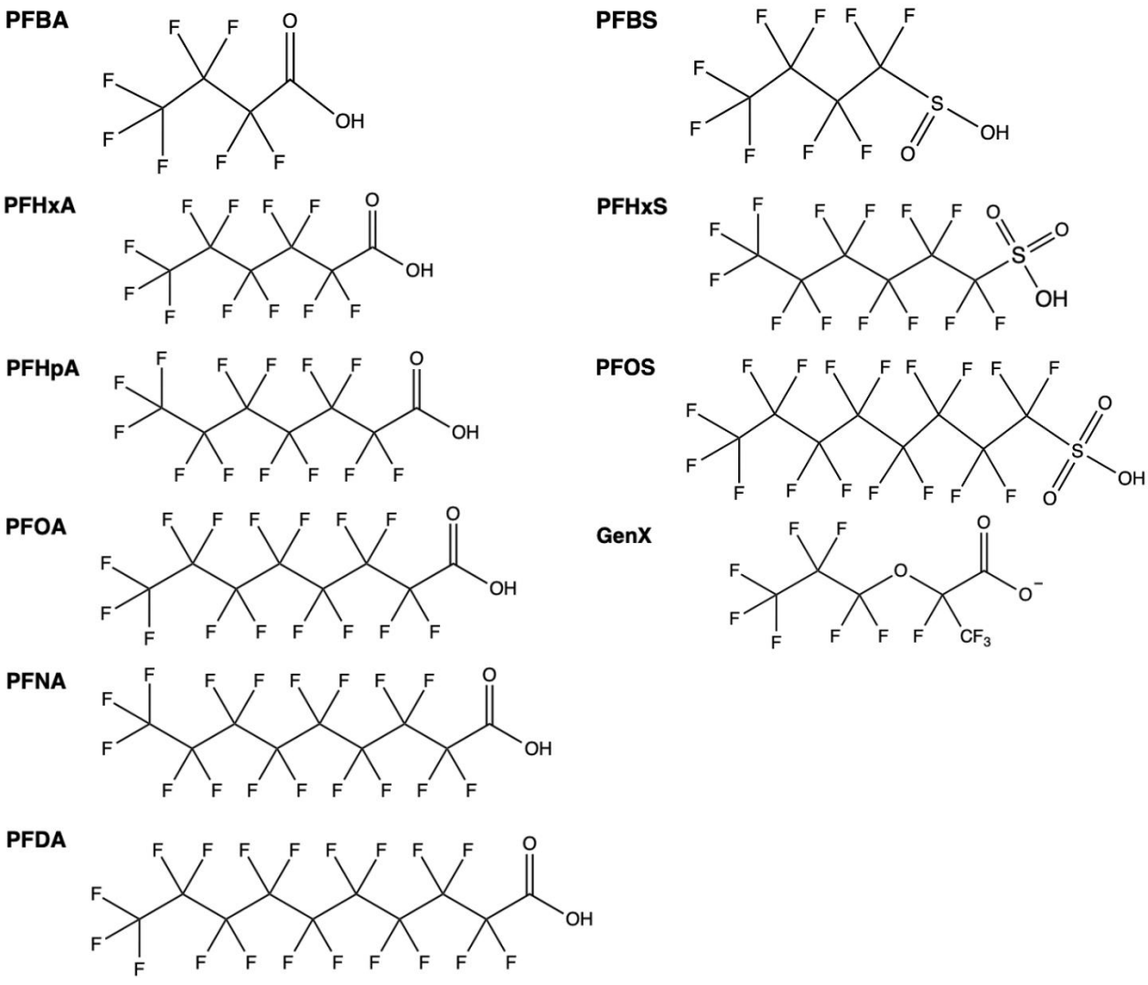


Figure S4. The structure of studied PFAS.

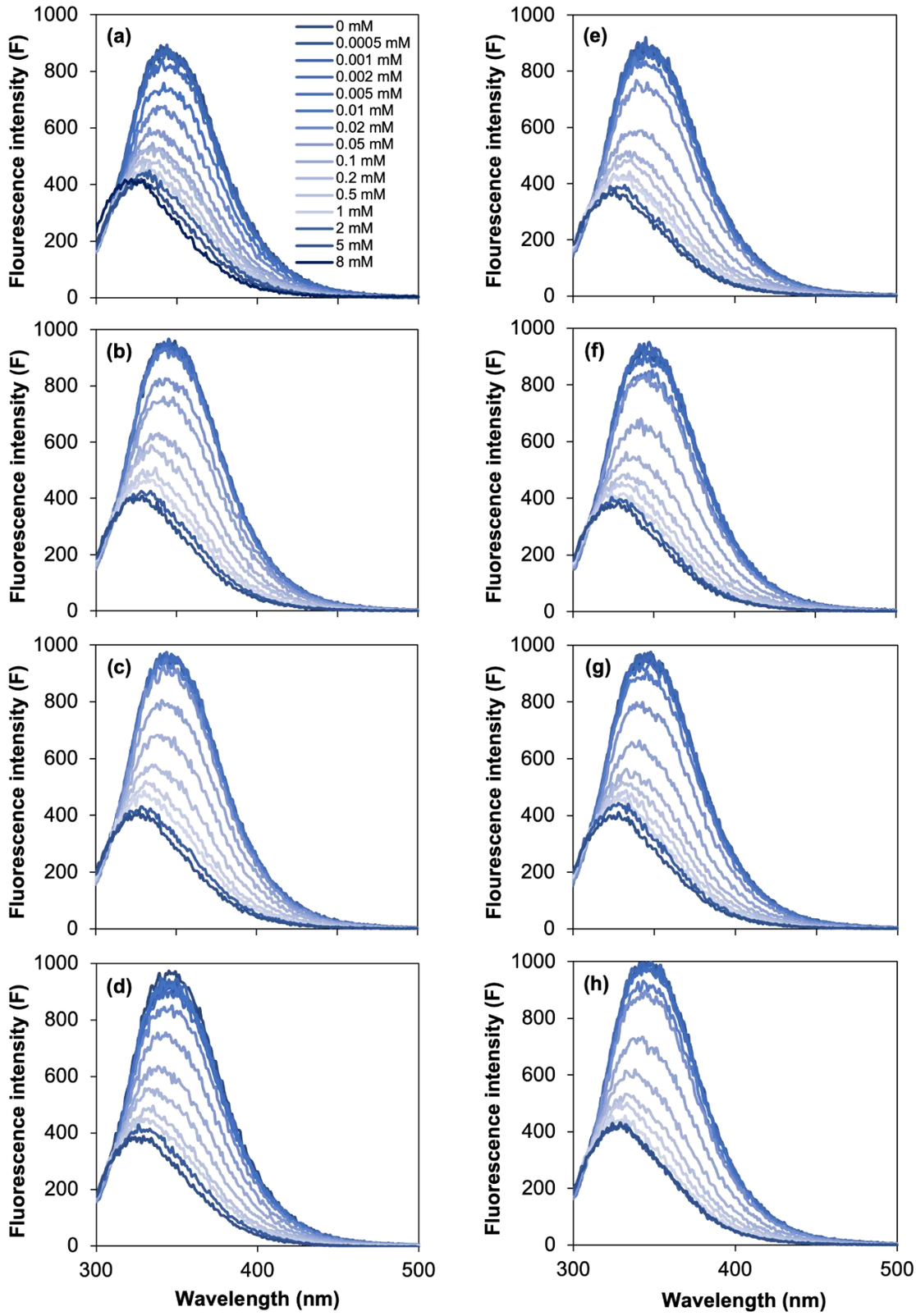


Figure S5. Fluorescence emission profiles of 0.8 μM BSA with increasing concentration of PFHpA and (a) in the absence of salt, and in the presence of (b) 150 mM NaCl, (c) 300 mM NaCl, (d) 150 mM KCl, (e) 50 mM CaCl2, (f) 150 mM, and (h) 150 mM MgCl2 in 20 mM MOPs buffer (pH 7) at 20 °C.

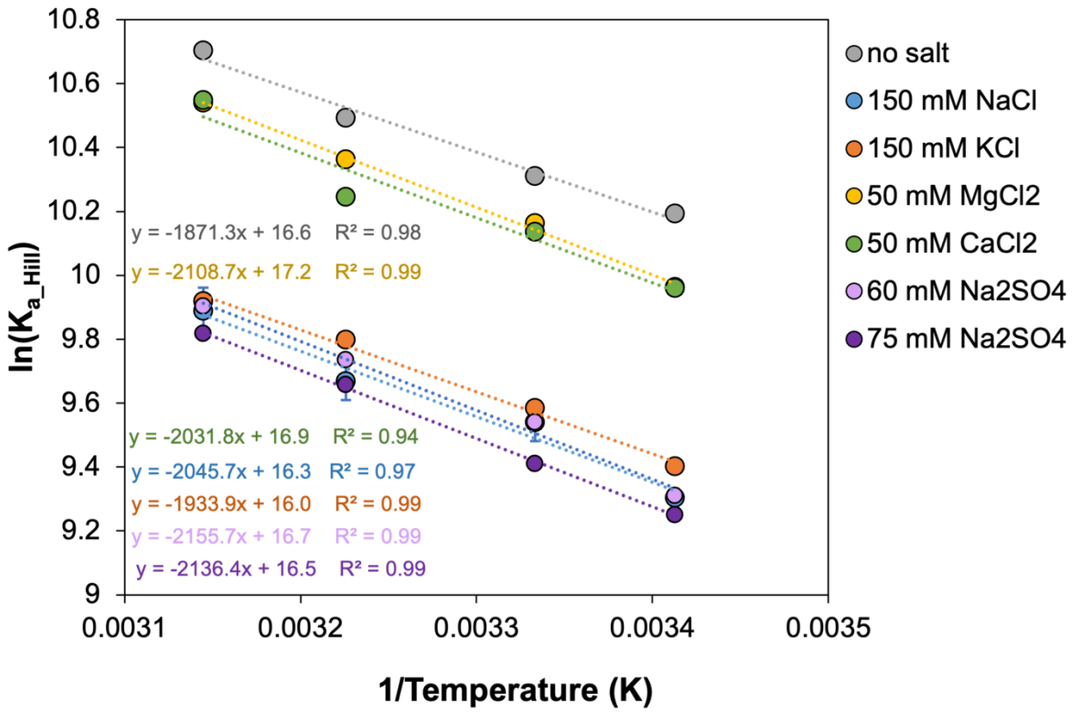


Figure S6. Van 't Hoff plot of 0.8 μM BSA with increasing concentration of PFHpA in the presence of different salts at (i.e., NaCl, KCl, $CaCl_2$, $MgCl_2$, and Na_2SO_4) in 20 mM MOPs solution at 20, 27, 37, and 45 °C. Markers are experimentally derived binding affinities. Dashed lines represent fitting to the Van 't Hoff equation as shown in Eq. (4).

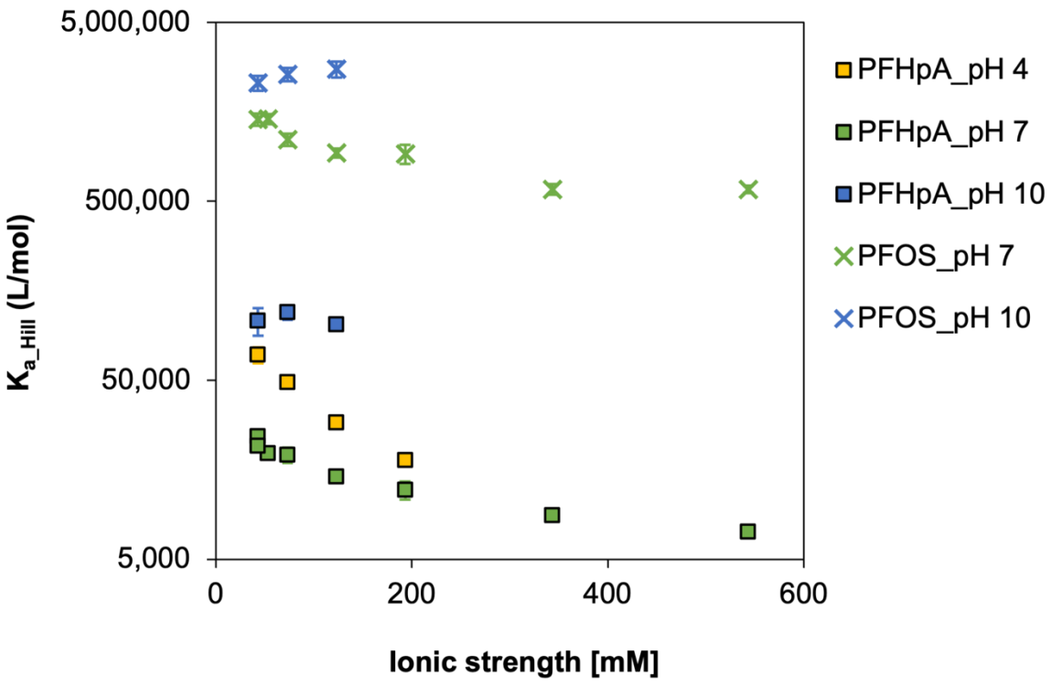


Figure S7. Experimentally measured binding constant of PFHpA and PFOS with BSA in the presence of NaCl salt in 20 mM citrate buffer at pH 4, 20 mM phosphate buffer at pH 7, and 20 mM phosphate buffer titrated with NaOH to pH 10. Error bars represent the standard error of the fit.

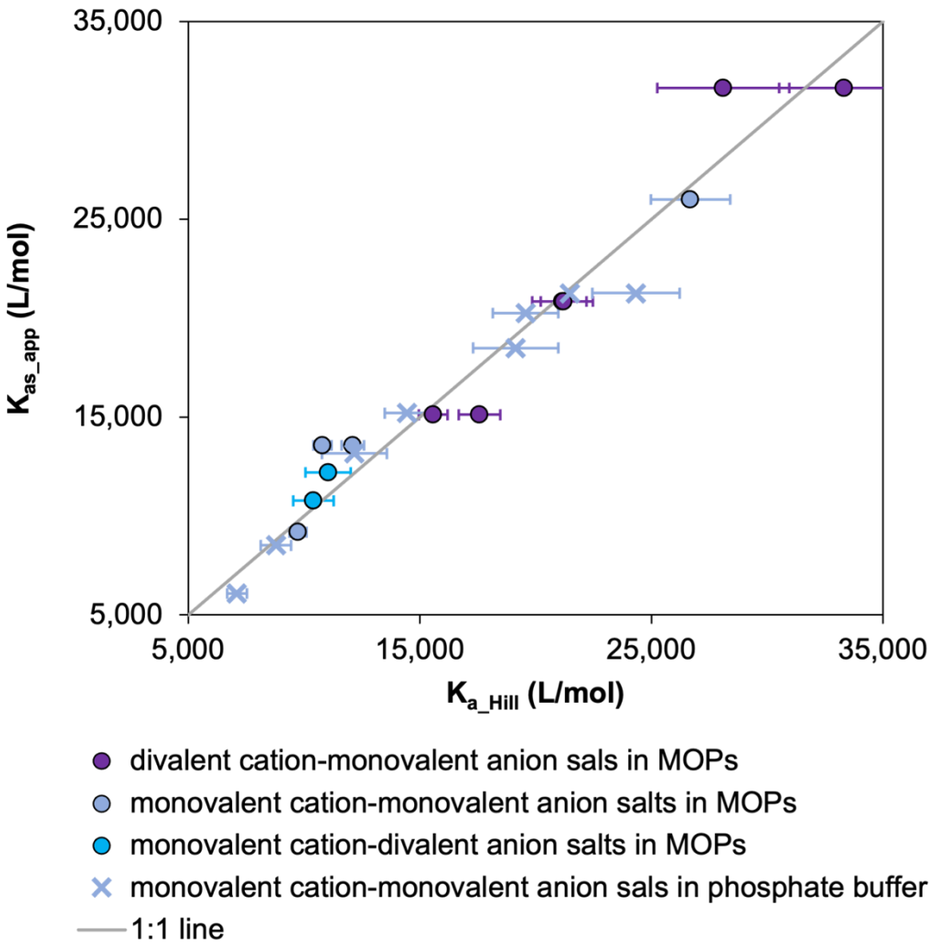


Figure S8. Model derived (K_{as_app}) and experimentally obtained (K_{a_Hill}) binding constants of PFHpA/BSA in the presence of different salts (i.e., NaCl, KCl, $CaCl_2$, $MgCl_2$, and Na_2SO_4 in MOPs or phosphate buffer solution at pH 7), with an R^2 value of 0.94.

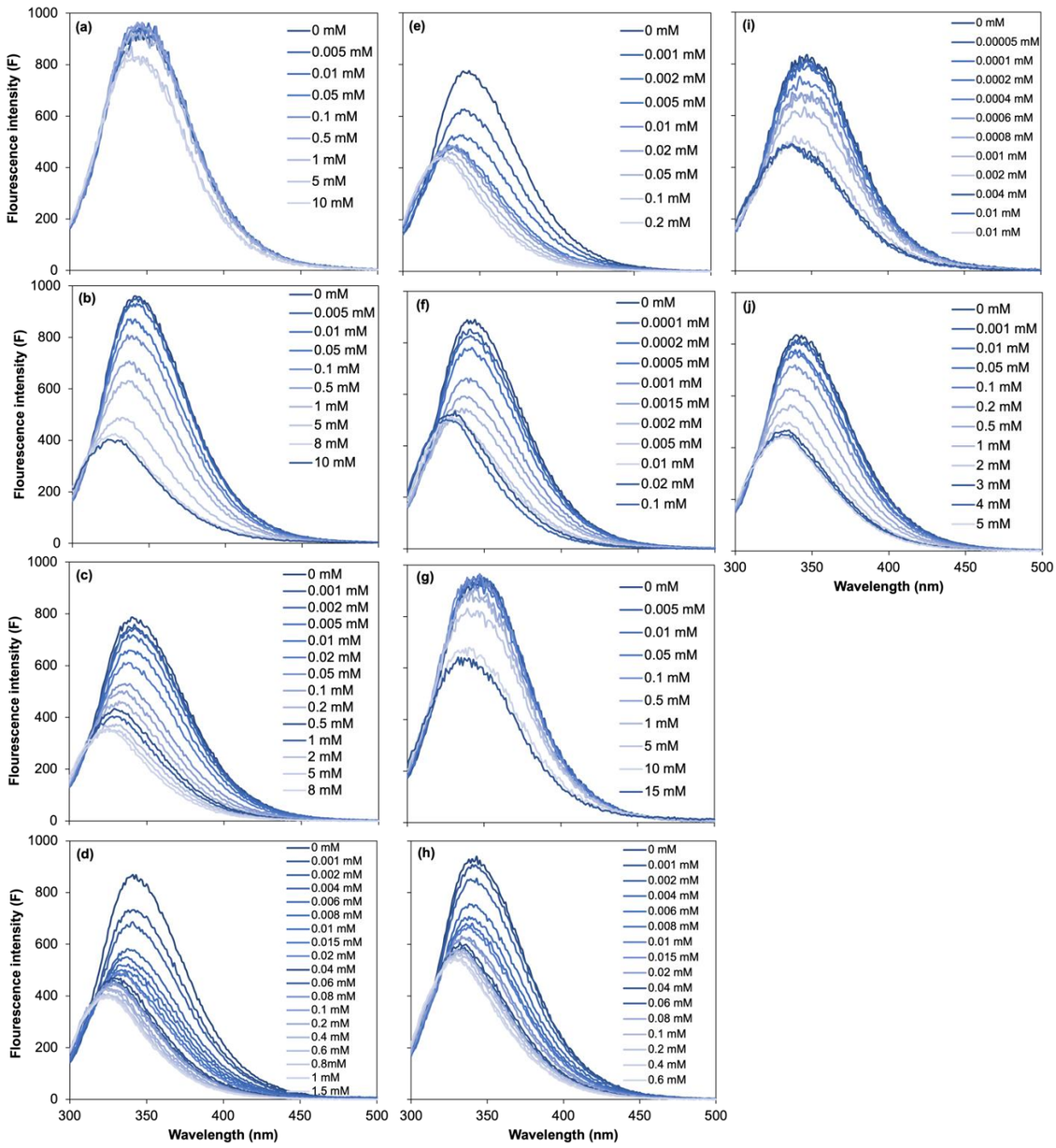


Figure S9. Fluorescence emission profiles of 0.8 μM BSA with increasing concentration of (a) PFBA, (b) PFHxA, (c) PFHpA, (d) PFOA, (e) PFNA, (f) PFDA, (g) PFBS, (h) PFHxS, (i) PFOS, and (j) GenX in the absence of salt in 20 mM NaH_2PO_4-Na_2HPO_4 buffer (pH 7) at 20 °C.

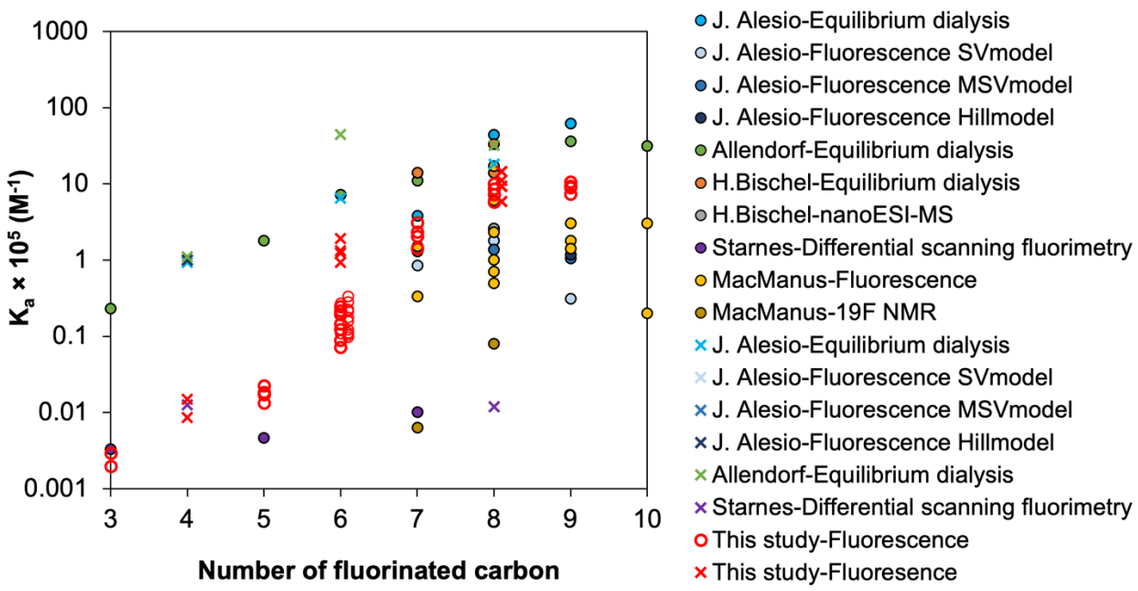


Figure S10. Comparison of binding constant (K_a), between literature studies (with experimental conditions listed in Table S1) and the present study for PFAS interactions with BSA under a range of salt conditions. Circle symbols represent PFCAs, while cross symbols represent PFSAs.

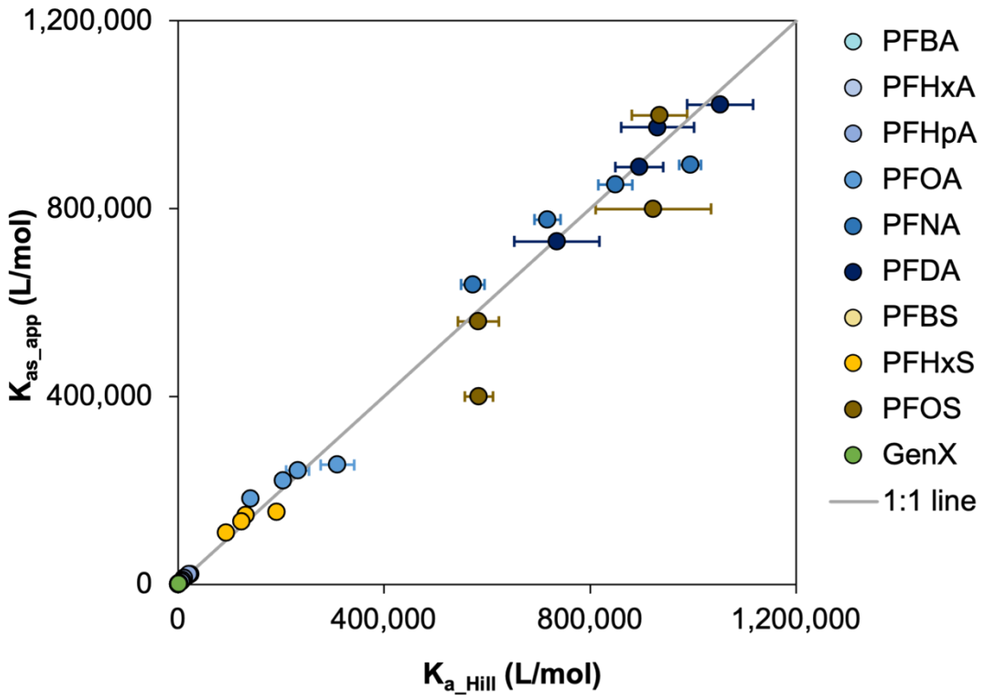


Figure S11. Model derived (K_{as_app}) and experimentally obtained (K_{a_Hill}) binding constants of PFAS/BSA in the presence of NaCl salts in 20 mM phosphate buffer solution at pH

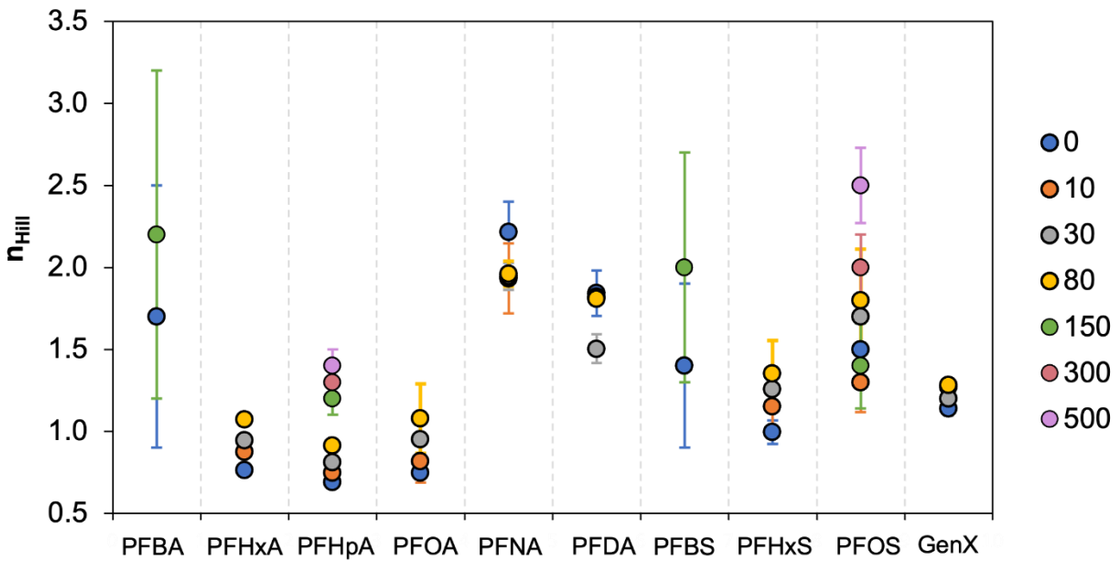


Figure S12. Experimentally obtained Hill coefficient (n_{Hil} from Eq. 7) of PFAS/BSA in 20 mM sodium phosphate buffer (pH 7, 20°C) in the absence and in the presence of 10, 30, 80, 150, 300 and 500 mM NaCl salt.

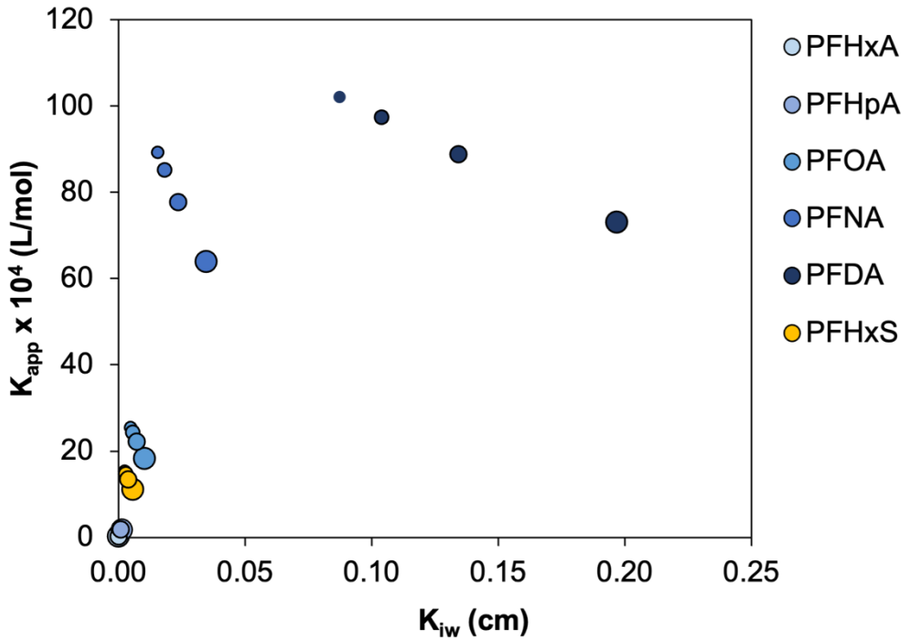


Figure S13. Model derived binding constants (K_{as_app}) and the interfacial adsorption distribution coefficient at low-PFAS-concentration (K_{iw_0}). The concentrations of salt used in predicted K_{iw} is based on the experimental condition of PFAS/BSA, with 32, 42, 62, and 112 mM NaCl. The size of symbols increases with salt concentration

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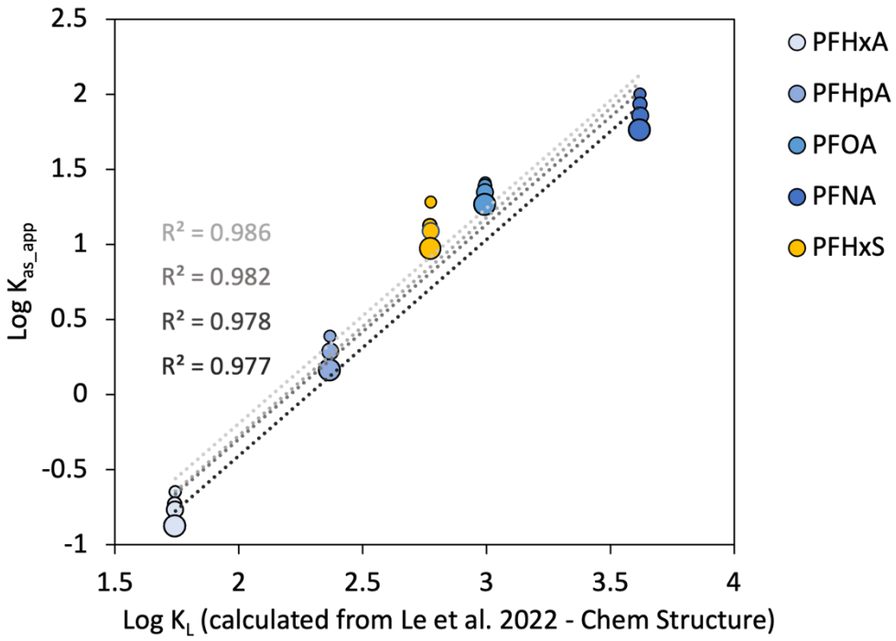


Figure S14. Log K_L (derived from Le, et al. [10]) vs Log K_{as_app} under different salt concentration for PFHxA, PFHpA, PFOA, and PFNA and PFHxS. The size of symbols increases with salt concentration increase from 32, to 112 mM (NaCl).

The parameter K_L in Le, et al. [10] study account for air-water adsorption of PFAS in the absence of additional salt, which is an indicator of hydrophobicity of PFAS. Fig. S8 shows a linear relation between Log K_L and Log K_{as_app} under all salt conditions indicating that the hydrophobic interaction (which increased with chain length) is the dominant driven force for PFAS/BSA interaction. Interestingly, PFHxS falls into the same fitting lines with PFCAs. Given the K_L values are based on chemical structure, which is easy to calculate/obtain. More work is required to determine the relation between Log K_L and Log K_{as_app} to explore if K_{as_app} can be predicted based on the K_L values.

Tables

Table S1. Binding constants of PFAS/protein from literature studies (ND: non-detected binding).

Fluorinated carbons	Ligand	Ligand concentration	Receptor	Receptor concentration	Buffer pH	Buffer	Buffer concentration (mol/L)	Ka $\times 10^{-5}$ (mol/L)_	Method	Reference
7	PFOA	0.001-0.02 mM	BSA	0.01 mM	7.4	PBS	1xPBS	3.77	Equilibrium dialysis	Alesio, et al. [11]
8	PFNA	0.001-0.02 mM	BSA	0.01 mM	7.4	PBS	1xPBS	44.05	Equilibrium dialysis	Alesio, et al. [11]
9	PFDA	0.001-0.02 mM	BSA	0.01 mM	7.4	PBS	1xPBS	61.62	Equilibrium dialysis	Alesio, et al. [11]
4	PFBS	0.001-0.02 mM	BSA	0.01 mM	7.4	PBS	1xPBS	0.93	Equilibrium dialysis	Alesio, et al. [11]
6	PFHxS	0.001-0.02 mM	BSA	0.01 mM	7.4	PBS	1xPBS	6.48	Equilibrium dialysis	Alesio, et al. [11]
8	PFOS	0.001-0.02 mM	BSA	0.01 mM	7.4	PBS	1xPBS	17.99	Equilibrium dialysis	Alesio, et al. [11]
5	GenX	0.001-0.02 mM	BSA	0.01 mM	7.4	PBS	1xPBS	0.65	Equilibrium dialysis	Alesio, et al. [11]
7	PFOA		BSA	0.00125 mM	7.4	PBS	1xPBS	0.85	Fluorescence-SV model	Alesio, et al. [11]
8	PFNA		BSA	0.00125 mM	7.4	PBS	1xPBS	1.78	Fluorescence-SV model	Alesio, et al. [11]
9	PFDA		BSA	0.00125 mM	7.4	PBS	1xPBS	0.31	Fluorescence-SV model	Alesio, et al. [11]
4	PFBS		BSA	0.00125 mM	7.4	PBS	1xPBS	0.21	Fluorescence-SV model	Alesio, et al. [11]
6	PFHxS		BSA	0.00125 mM	7.4	PBS	1xPBS	0.48	Fluorescence-SV model	Alesio, et al. [11]
8	PFOS		BSA	0.00125 mM	7.4	PBS	1xPBS	1.34	Fluorescence-SV model	Alesio, et al. [11]
5	GenX		BSA	0.00125 mM	7.4	PBS	1xPBS	0.24	Fluorescence-SV model	Alesio, et al. [11]
7	PFOA		BSA	0.00125 mM	7.4	PBS	1xPBS	1.41	Fluorescence-MSV model	Alesio, et al. [11]
8	PFNA		BSA	0.00125 mM	7.4	PBS	1xPBS	1.37	Fluorescence-MSV model	Alesio, et al. [11]
9	PFDA		BSA	0.00125 mM	7.4	PBS	1xPBS	1.04	Fluorescence-MSV model	Alesio, et al. [11]
4	PFBS		BSA	0.00125 mM	7.4	PBS	1xPBS	0.2	Fluorescence-MSV model	Alesio, et al. [11]
6	PFHxS		BSA	0.00125 mM	7.4	PBS	1xPBS	0.8	Fluorescence-MSV model	Alesio, et al. [11]
8	PFOS		BSA	0.00125 mM	7.4	PBS	1xPBS	1.33	Fluorescence-MSV model	Alesio, et al. [11]
5	GenX		BSA	0.00125 mM	7.4	PBS	1xPBS	0.1	Fluorescence-MSV model	Alesio, et al. [11]
7	PFOA		BSA	0.00125 mM	7.4	PBS	1xPBS	1.39	Fluorescence-Hill model	Alesio, et al. [11]
8	PFNA		BSA	0.00125 mM	7.4	PBS	1xPBS	2.55	Fluorescence-Hill model	Alesio, et al. [11]
9	PFDA		BSA	0.00125 mM	7.4	PBS	1xPBS	1.17	Fluorescence-Hill model	Alesio, et al. [11]
4	PFBS		BSA	0.00125 mM	7.4	PBS	1xPBS	0.2	Fluorescence-Hill model	Alesio, et al. [11]
6	PFHxS		BSA	0.00125 mM	7.4	PBS	1xPBS	0.83	Fluorescence-Hill model	Alesio, et al. [11]
8	PFOS		BSA	0.00125 mM	7.4	PBS	1xPBS	1.14	Fluorescence-Hill model	Alesio, et al. [11]

5	GenX		BSA	0.00125 mM	7.4	PBS	1xPBS	0.14	Fluorescence-Hill model	Alesio, et al. [11]
3	PFBA		BSA	3 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	0.23	Dialysis experiment	Allendorf, et al. [2]
5	PFHxA		BSA	1 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	1.8	Dialysis experiment	Allendorf, et al. [2]
6	PFHpA		BSA	0.1 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	7.2	Dialysis experiment	Allendorf, et al. [2]
7	PFOA		BSA	0.1 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	11	Dialysis experiment	Allendorf, et al. [2]
8	PFNA		BSA	0.05 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	14	Dialysis experiment	Allendorf, et al. [2]
9	PFDA		BSA	0.05 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	36	Dialysis experiment	Allendorf, et al. [2]
10	PFUnA		BSA	0.025 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	31	Dialysis experiment	Allendorf, et al. [2]
4	PFBS		BSA	1 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	1.1	Dialysis experiment	Allendorf, et al. [2]
6	PFHxS		BSA	0.05 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	44	Dialysis experiment	Allendorf, et al. [2]
8	PFOS		BSA	0.025 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	32	Dialysis experiment	Allendorf, et al. [2]
5	GenX		BSA	3 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	0.23	Dialysis experiment	Allendorf, et al. [2]
	DONA		BSA	0.1 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	1.8	Dialysis experiment	Allendorf, et al. [2]
	9CL-PF3ONS		BSA	0.025 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	69	Dialysis experiment	Allendorf, et al. [2]
	PFECHS		BSA	0.1 ng/L	7.4	Hank's balanced salt solution (HBSS)	1xHBSS with 1.86 g/L additional salts	24	Dialysis experiment	Allendorf, et al. [2]
6	PFHpA		HAS	0.004 mM	7	phosphate buffer	50 mM	0.09	Fluorescence	Hebert and MacManus-Spencer [4]
7	PFOA		HAS	0.004 mM	7	phosphate buffer	50 mM	0.22	Fluorescence	Hebert and MacManus-Spencer [4]
7	PFOA		HAS	0.01 mM	7	phosphate buffer	50 mM	0.13	Fluorescence	Hebert and MacManus-Spencer [4]

8	PFNA		HAS	0.004 mM	7	phosphate buffer	50 mM	0.26	Fluorescence	Hebert and MacManus-Spencer [4]
8	PFNA		HAS	0.01 mM	7	phosphate buffer	50 mM	0.10	Fluorescence	Hebert and MacManus-Spencer [4]
9	PFDA		HAS	0.004 mM	7	phosphate buffer	50 mM	0.47	Fluorescence	Hebert and MacManus-Spencer [4]
10	PFUnA		HAS	0.004 mM	7	phosphate buffer	50 mM	0.42	Fluorescence	Hebert and MacManus-Spencer [4]
6	PFHxS		HAS	0.004 mM	7	phosphate buffer	50 mM	0.12	Fluorescence	Hebert and MacManus-Spencer [4]
8	PFOS		HAS	0.004 mM	7	phosphate buffer	50 mM	0.09	Fluorescence	Hebert and MacManus-Spencer [4]
7	PFOA	0.04-70 ratio	BSA	0.5 mM	7.4	sodium phosphate buffer	50 mM	14	dialysis	Bischel, et al. [3]
7	PFOA	0.1-4 ratio	BSA	0.5 mM	7.4	sodium phosphate buffer	50 mM	1.3	nanoESI-MS	Bischel, et al. [3]
8	PFNA	0.0005 ratio	BSA	0.5 mM	7.4	sodium phosphate buffer	50 mM	14	dialysis	Bischel, et al. [3]
8	PFNA	0.02-120 ratio	BSA	0.5 mM	7.4	sodium phosphate buffer	50 mM	33	dialysis	Bischel, et al. [3]
8	PFNA	0.1-4 ratio	BSA	0.5 mM	7.4	sodium phosphate buffer	50 mM	2.6	nanoESI-MS	Bischel, et al. [3]
8	PFNA	0.0006 ratio	HAS	0.5 mM	7.4	sodium phosphate buffer	50 mM	21	dialysis	Bischel, et al. [3]
3	PFBA		HSA		7.4	NaH2PO4-Na2HPO4	100 mM	nd	fluorescence	Chen and Guo [7]
7	PFOA		HSA		7.4	NaH2PO4-Na2HPO4	100 mM	2.7	fluorescence	Chen and Guo [7]
11	PFDoA		HSA		7.4	NaH2PO4-Na2HPO4	100 mM	nd	fluorescence	Chen and Guo [7]
4	PFBS		HSA		7.4	NaH2PO4-Na2HPO4	100 mM	nd	fluorescence	Chen and Guo [7]
8	PFOS		HSA		7.4	NaH2PO4-Na2HPO4	100 mM	0.22	fluorescence	Chen and Guo [7]
8	PFOS		HSA					125	nanoESI-MS	MacManus-Spencer, et al. [6]
3	PFBA		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0038	Differential Scanning Fluorimetry	Starnes, et al. [5]
5	PFHxA		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0058	Differential Scanning Fluorimetry	Starnes, et al. [5]
7	PFOA		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0127	Differential Scanning Fluorimetry	Starnes, et al. [5]
4	PFBS		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0060	Differential Scanning Fluorimetry	Starnes, et al. [5]
8	PFOS		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0145	Differential Scanning Fluorimetry	Starnes, et al. [5]
5	GenX		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0064	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTOH		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	nd	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTSA		HSA	0.13 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0244	Differential Scanning Fluorimetry	Starnes, et al. [5]
3	PFBA		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0033	Differential Scanning Fluorimetry	Starnes, et al. [5]

5	PFHxA		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0046	Differential Scanning Fluorimetry	Starnes, et al. [5]
7	PFOA		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0101	Differential Scanning Fluorimetry	Starnes, et al. [5]
4	PFBS		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0125	Differential Scanning Fluorimetry	Starnes, et al. [5]
8	PFOS		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0118	Differential Scanning Fluorimetry	Starnes, et al. [5]
5	GenX		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0039	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTOH		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	nd	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTSA		BSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0072	Differential Scanning Fluorimetry	Starnes, et al. [5]
3	PFBA		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0078	Differential Scanning Fluorimetry	Starnes, et al. [5]
5	PFHxA		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0105	Differential Scanning Fluorimetry	Starnes, et al. [5]
7	PFOA		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0270	Differential Scanning Fluorimetry	Starnes, et al. [5]
4	PFBS		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0080	Differential Scanning Fluorimetry	Starnes, et al. [5]
8	PFOS		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0263	Differential Scanning Fluorimetry	Starnes, et al. [5]
5	GenX		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0047	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTOH		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	nd	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTSA		PSA	0.11 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0192	Differential Scanning Fluorimetry	Starnes, et al. [5]
3	PFBA		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0053	Differential Scanning Fluorimetry	Starnes, et al. [5]
5	PFHxA		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0075	Differential Scanning Fluorimetry	Starnes, et al. [5]
7	PFOA		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0250	Differential Scanning Fluorimetry	Starnes, et al. [5]
4	PFBS		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0071	Differential Scanning Fluorimetry	Starnes, et al. [5]
8	PFOS		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0238	Differential Scanning Fluorimetry	Starnes, et al. [5]
5	GenX		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0061	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTOH		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	nd	Differential Scanning Fluorimetry	Starnes, et al. [5]
6	6:2 FTSA		RSA	0.18 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.0303	Differential Scanning Fluorimetry	Starnes, et al. [5]
3	PFBA		HSA	0.006 mM	7.4	PBS	0.1 M	0.024	Equilibrium dialysis	Gao, et al. [12]
4	PFPeA		HSA	0.006 mM	7.4	PBS	0.1 M	0.035	Equilibrium dialysis	Gao, et al. [12]
5	PFHxA		HSA	0.006 mM	7.4	PBS	0.1 M	0.051	Equilibrium dialysis	Gao, et al. [12]
6	PFHpA		HSA	0.006 mM	7.4	PBS	0.1 M	0.058	Equilibrium dialysis	Gao, et al. [12]
7	PFOA		HSA	0.006 mM	7.4	PBS	0.1 M	0.087	Equilibrium dialysis	Gao, et al. [12]
8	PFNA		HSA	0.006 mM	7.4	PBS	0.1 M	0.137	Equilibrium dialysis	Gao, et al. [12]
9	PFDA		HSA	0.006 mM	7.4	PBS	0.1 M	0.154	Equilibrium dialysis	Gao, et al. [12]

10	PFUnDA		HSA	0.006 mM	7.4	PBS	0.1 M	0.127	Equilibrium dialysis	Gao, et al. [12]
11	PFDoA		HSA	0.006 mM	7.4	PBS	0.1 M	0.097	Equilibrium dialysis	Gao, et al. [12]
12	PFTrDA		HSA	0.006 mM	7.4	PBS	0.1 M	0.038	Equilibrium dialysis	Gao, et al. [12]
4	PFBS		HSA	0.006 mM	7.4	PBS	0.1 M	0.059	Equilibrium dialysis	Gao, et al. [12]
6	PFHxS		HSA	0.006 mM	7.4	PBS	0.1 M	0.127	Equilibrium dialysis	Gao, et al. [12]
8	PFOS		HSA	0.006 mM	7.4	PBS	0.1 M	0.263	Equilibrium dialysis	Gao, et al. [12]
6	6:2Cl-PFESA		HSA	0.006 mM	7.4	PBS	0.1 M	0.149	Equilibrium dialysis	Gao, et al. [12]
3	PFBA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.011	Equilibrium dialysis	Gao, et al. [12]
4	PFPeA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.015	Equilibrium dialysis	Gao, et al. [12]
5	PFHxA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.023	Equilibrium dialysis	Gao, et al. [12]
6	PFHpA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.028	Equilibrium dialysis	Gao, et al. [12]
7	PFOA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.051	Equilibrium dialysis	Gao, et al. [12]
8	PFNA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.111	Equilibrium dialysis	Gao, et al. [12]
9	PFDA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.130	Equilibrium dialysis	Gao, et al. [12]
10	PFUnDA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.185	Equilibrium dialysis	Gao, et al. [12]
11	PFDoA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.105	Equilibrium dialysis	Gao, et al. [12]
12	PFTrDA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.032	Equilibrium dialysis	Gao, et al. [12]
4	PFBS		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.023	Equilibrium dialysis	Gao, et al. [12]
6	PFHxS		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.048	Equilibrium dialysis	Gao, et al. [12]
8	PFOS		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.123	Equilibrium dialysis	Gao, et al. [12]
6	6:2Cl-PFESA		L-FABP	0.006 mM	7.4	PBS	0.1 M	0.098	Equilibrium dialysis	Gao, et al. [12]
8	PFOS		HSA	0.006 mM		no buffer		125.000	Ultrafiltration	Beesoon and Martin [8]
8	3m-PFOS		HSA	0.006 mM		no buffer		0.025	Ultrafiltration	Beesoon and Martin [8]
8	4m-PFOS		HSA	0.006 mM		no buffer		0.125	Ultrafiltration	Beesoon and Martin [8]
8	5m-PFOS		HSA	0.006 mM		no buffer		0.111	Ultrafiltration	Beesoon and Martin [8]
7	PFOA		HSA	0.006 mM		no buffer		0.100	Ultrafiltration	Beesoon and Martin [8]
7	3m-PFOA		HSA	0.006 mM		no buffer		0.025	Ultrafiltration	Beesoon and Martin [8]
7	5m-PFOA		HSA	0.006 mM		no buffer		0.033	Ultrafiltration	Beesoon and Martin [8]
3	PFBA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.004	Differential Scanning Fluorimetry	Jackson, et al. [9]

4	PFPeA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.005	Differential Scanning Fluorimetry	Jackson, et al. [9]
5	PFHxA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.006	Differential Scanning Fluorimetry	Jackson, et al. [9]
6	PFHpA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.023	Differential Scanning Fluorimetry	Jackson, et al. [9]
7	PFOA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.012	Differential Scanning Fluorimetry	Jackson, et al. [9]
8	PFNA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.017	Differential Scanning Fluorimetry	Jackson, et al. [9]
9	PFDA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.008	Differential Scanning Fluorimetry	Jackson, et al. [9]
10	PFUnDA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.007	Differential Scanning Fluorimetry	Jackson, et al. [9]
11	PFDoDA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.005	Differential Scanning Fluorimetry	Jackson, et al. [9]
4	PFBS		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.006	Differential Scanning Fluorimetry	Jackson, et al. [9]
6	PFHxS		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.014	Differential Scanning Fluorimetry	Jackson, et al. [9]
8	PFOS		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.014	Differential Scanning Fluorimetry	Jackson, et al. [9]
5	GenX		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.006	Differential Scanning Fluorimetry	Jackson, et al. [9]
4	4:2FTOH		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	nd	Differential Scanning Fluorimetry	Jackson, et al. [9]
6	6:2FTOH		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	nd	Differential Scanning Fluorimetry	Jackson, et al. [9]
3	3:3FTCA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.006	Differential Scanning Fluorimetry	Jackson, et al. [9]
5	5:3FTCA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.016	Differential Scanning Fluorimetry	Jackson, et al. [9]
6	6:3FTCA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.012	Differential Scanning Fluorimetry	Jackson, et al. [9]

8	8:3FTCA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.010	Differential Scanning Fluorimetry	Jackson, et al. [9]
4	4:2FTSA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.009	Differential Scanning Fluorimetry	Jackson, et al. [9]
6	6:2FTSA		HSA	0.125 mM	7.4	HEPES buffered saline	140 mM NaCl, 50 mM HEPES, 0.38 mM Na2HPO4	0.027	Differential Scanning Fluorimetry	Jackson, et al. [9]
6	PFHpA	298 K	HSA	0.001 mM	7.4	PBS		0.05	fluorescence	Alesio and Bothun [1]
6	PFHpA	304 K	HSA	0.001 mM	7.4	PBS		5.35	fluorescence	Alesio and Bothun [1]
6	PFHpA	310 K	HSA	0.001 mM	7.4	PBS		7.47	fluorescence	Alesio and Bothun [1]
7	PFOA	298 K	HSA	0.001 mM	7.4	PBS		2.27	fluorescence	Alesio and Bothun [1]
7	PFOA	304 K	HSA	0.001 mM	7.4	PBS		22.30	fluorescence	Alesio and Bothun [1]
7	PFOA	310 K	HSA	0.001 mM	7.4	PBS		19.80	fluorescence	Alesio and Bothun [1]
8	PFNA	298 K	HSA	0.001 mM	7.4	PBS		78.10	fluorescence	Alesio and Bothun [1]
8	PFNA	304 K	HSA	0.001 mM	7.4	PBS		2.02	fluorescence	Alesio and Bothun [1]
8	PFNA	310 K	HSA	0.001 mM	7.4	PBS		0.99	fluorescence	Alesio and Bothun [1]
8	HFPO-TA	298 K	HSA	0.001 mM	7.4	PBS		37.00	fluorescence	Alesio and Bothun [1]
8	HFPO-TA	304 K	HSA	0.001 mM	7.4	PBS		854.00	fluorescence	Alesio and Bothun [1]
8	HFPO-TA	310 K	HSA	0.001 mM	7.4	PBS		23100.00	fluorescence	Alesio and Bothun [1]
6	PFO3DA	298 K	HSA	0.001 mM	7.4	PBS		1.59	fluorescence	Alesio and Bothun [1]
6	PFO3DA	304 K	HSA	0.001 mM	7.4	PBS		0.46	fluorescence	Alesio and Bothun [1]
6	PFO3DA	310 K	HSA	0.001 mM	7.4	PBS		0.30	fluorescence	Alesio and Bothun [1]
6	DFSA	298 K	HSA	0.001 mM	7.4	PBS		0.02	fluorescence	Alesio and Bothun [1]
6	DFSA	304 K	HSA	0.001 mM	7.4	PBS		0.05	fluorescence	Alesio and Bothun [1]
6	DFSA	310 K	HSA	0.001 mM	7.4	PBS		0.21	fluorescence	Alesio and Bothun [1]
7	PFOA		BSA	0.01 mM	7.4	sodium phosphate buffer	50 mM	0.33	Fluorescence	MacManus-Spencer, et al. [6]
7	PFOA		BSA	0.001 mM	7.4	sodium phosphate buffer	50 mM	1.5	Fluorescence	MacManus-Spencer, et al. [6]
8	PFNA		BSA	0.01 mM	7.4	sodium phosphate buffer	50 mM	0.49	Fluorescence	MacManus-Spencer, et al. [6]
8	PFNA		BSA	0.008 mM	7.4	sodium phosphate buffer	50 mM	0.7	Fluorescence	MacManus-Spencer, et al. [6]
8	PFNA		BSA	0.006 mM	7.4	sodium phosphate buffer	50 mM	1	Fluorescence	MacManus-Spencer, et al. [6]

8	PFNA		BSA	0.004 mM	7.4	sodium phosphate buffer	50 mM	17	Fluorescence	MacManus-Spencer, et al. [6]
8	PFNA		BSA	0.002 mM	7.4	sodium phosphate buffer	50 mM	2.3	Fluorescence	MacManus-Spencer, et al. [6]
8	PFNA		BSA	0.001 mM	7.4	sodium phosphate buffer	50 mM	6	Fluorescence	MacManus-Spencer, et al. [6]
9	PFDA		BSA	0.01 mM	7.4	sodium phosphate buffer	50 mM	1.8	Fluorescence	MacManus-Spencer, et al. [6]
9	PFDA		BSA	0.002 mM	7.4	sodium phosphate buffer	50 mM	1.4	Fluorescence	MacManus-Spencer, et al. [6]
9	PFDA		BSA	0.001 mM	7.4	sodium phosphate buffer	50 mM	3	Fluorescence	MacManus-Spencer, et al. [6]
10	PFUnA		BSA	0.01 mM	7.4	sodium phosphate buffer	50 mM	3	Fluorescence	MacManus-Spencer, et al. [6]
10	PFUnA		BSA	0.001 mM	7.4	sodium phosphate buffer	50 mM	0.2	Fluorescence	MacManus-Spencer, et al. [6]
8	PFNA		HSA	0.01 mM	7.4	sodium phosphate buffer	50 mM	0.3	Fluorescence	MacManus-Spencer, et al. [6]
8	PFNA		HSA	0.001 mM	7.4	sodium phosphate buffer	50 mM	0.5	Fluorescence	MacManus-Spencer, et al. [6]
7	PFOA		BSA		7.4	sodium phosphate buffer	50 mM	0.0063	19 F NMR	MacManus-Spencer, et al. [6]
8	PFNA		BSA		7.4	sodium phosphate buffer	50 mM	0.08	19 F NMR	MacManus-Spencer, et al. [6]
8	PFNA		HSA		7.4	sodium phosphate buffer	50 mM	0.2	19 F NMR	MacManus-Spencer, et al. [6]
5	PFHxA	1-1 ratio	L-FABP		7.4	PBS	1xPBS	nd	Equilibrium Dialysis	Khazaee, et al. [13]
7	PFOA	1-1 ratio	L-FABP		7.4	PBS	1xPBS	101.01	Equilibrium Dialysis	Khazaee, et al. [13]
4	PFBS	1-1 ratio	L-FABP		7.4	PBS	1xPBS	nd	Equilibrium Dialysis	Khazaee, et al. [13]
8	PFOS	1-1 ratio	L-FABP		7.4	PBS	1xPBS	54.35	Equilibrium Dialysis	Khazaee, et al. [13]
6	PFHpA	1-1 ratio	I-FABP		7.4	PBS	1xPBS	nd	Equilibrium Dialysis	Khazaee, et al. [13]
8	PFNA	1-1 ratio	I-FABP		7.4	PBS	1xPBS	nd	Equilibrium Dialysis	Khazaee, et al. [13]
3	PFBA	1-1 ratio	PPAR-α		7.4	PBS	1xPBS	nd	Equilibrium Dialysis	Khazaee, et al. [13]
5	PFHxA	1-1 ratio	PPAR-α		7.4	PBS	1xPBS	103.1	Equilibrium Dialysis	Khazaee, et al. [13]
6	PFHpA	1-1 ratio	PPAR-α		7.4	PBS	1xPBS	nd	Equilibrium Dialysis	Khazaee, et al. [13]
8	PFNA	1-1 ratio	PPAR-α		7.4	PBS	1xPBS	120.48	Equilibrium Dialysis	Khazaee, et al. [13]
7	PFOA	1-1 ratio	PPAR-γ		7.4	PBS	1xPBS	175.44	Equilibrium Dialysis	Khazaee, et al. [13]
8	PFOS	1-1 ratio	PPAR-γ		7.4	PBS	1xPBS	1.18	Equilibrium Dialysis	Khazaee, et al. [13]

3	PFBA	1-1 ratio	PPAR-δ		7.4	PBS	1xPBS	227.27	Equilibrium Dialysis	Khazaee, et al. [13]
4	PFBS	1-1 ratio	PPAR-δ		7.4	PBS	1xPBS	nd	Equilibrium Dialysis	Khazaee, et al. [13]
6	PFHxS	1-1 ratio	PPAR-δ		7.4	PBS	1xPBS	285.71	Equilibrium Dialysis	Khazaee, et al. [13]
8	PFOS	1-1 ratio	PPAR-δ		7.4	PBS	1xPBS	1.50	Equilibrium Dialysis	Khazaee, et al. [13]
3	PFBA		L-FABP	0.0004 mM	8	Tris- HCl buffer		nd	Fluorescence Displacement	Zhang, et al. [14]
4	PFPeA		L-FABP	0.0004 mM	8	Tris- HCl buffer		nd	Fluorescence Displacement	Zhang, et al. [14]
5	PFHxA		L-FABP	0.0004 mM	8	Tris- HCl buffer		nd	Fluorescence Displacement	Zhang, et al. [14]
6	PFHpA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.03	Fluorescence Displacement	Zhang, et al. [14]
7	PFOA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.20	Fluorescence Displacement	Zhang, et al. [14]
8	PFNA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.62	Fluorescence Displacement	Zhang, et al. [14]
9	PFDA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.78	Fluorescence Displacement	Zhang, et al. [14]
10	PFUnDA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.94	Fluorescence Displacement	Zhang, et al. [14]
11	PFDoDA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.81	Fluorescence Displacement	Zhang, et al. [14]
13	PFTeDA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.17	Fluorescence Displacement	Zhang, et al. [14]
15	PFHxDA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.09	Fluorescence Displacement	Zhang, et al. [14]
17	PFOcDA		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.16	Fluorescence Displacement	Zhang, et al. [14]
4	PFBS		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.01	Fluorescence Displacement	Zhang, et al. [14]
6	PFHxS		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.12	Fluorescence Displacement	Zhang, et al. [14]
8	PFOS		L-FABP	0.0004 mM	8	Tris- HCl buffer		0.54	Fluorescence Displacement	Zhang, et al. [14]
6	6:2 FTOH		L-FABP	0.0004 mM	8	Tris- HCl buffer		nd	Fluorescence Displacement	Zhang, et al. [14]
8	8:2 FTOH		L-FABP	0.0004 mM	8	Tris- HCl buffer		nd	Fluorescence Displacement	Zhang, et al. [14]
3	PFBA		PPAR-γ	0.0008 mM	7.4	PBS		nd	Fluorescence Displacement	Zhang, et al. [15]
5	PFHxA		PPAR-γ	0.0008 mM	7.4	PBS		nd	Fluorescence Displacement	Zhang, et al. [15]
6	PFHpA		PPAR-γ	0.0008 mM	7.4	PBS		0.008	Fluorescence Displacement	Zhang, et al. [15]
7	PFOA		PPAR-γ	0.0008 mM	7.4	PBS		0.033	Fluorescence Displacement	Zhang, et al. [15]
8	PFNA		PPAR-γ	0.0008 mM	7.4	PBS		0.064	Fluorescence Displacement	Zhang, et al. [15]
9	PFDA		PPAR-γ	0.0008 mM	7.4	PBS		0.118	Fluorescence Displacement	Zhang, et al. [15]
10	PFUnDA		PPAR-γ	0.0008 mM	7.4	PBS		0.172	Fluorescence Displacement	Zhang, et al. [15]
11	PFDoDA		PPAR-γ	0.0008 mM	7.4	PBS		0.070	Fluorescence Displacement	Zhang, et al. [15]
13	PFTeDA		PPAR-γ	0.0008 mM	7.4	PBS		0.063	Fluorescence Displacement	Zhang, et al. [15]
15	PFHxDA		PPAR-γ	0.0008 mM	7.4	PBS		0.078	Fluorescence Displacement	Zhang, et al. [15]

17	PFOcDA		PPAR-γ	0.0008 mM	7.4	PBS		0.093	Fluorescence Displacement	Zhang, et al. [15]
4	PFBS		PPAR-γ	0.0008 mM	7.4	PBS		nd	Fluorescence Displacement	Zhang, et al. [15]
6	PFHxS		PPAR-γ	0.0008 mM	7.4	PBS		0.035	Fluorescence Displacement	Zhang, et al. [15]
8	PFOS		PPAR-γ	0.0008 mM	7.4	PBS		0.107	Fluorescence Displacement	Zhang, et al. [15]
6	6:2 FTOH		PPAR-γ	0.0008 mM	7.4	PBS		nd	Fluorescence Displacement	Zhang, et al. [15]
8	8:2 FTOH		PPAR-γ	0.0008 mM	7.4	PBS		nd	Fluorescence Displacement	Zhang, et al. [15]

Table S2. Average percentages of equivalents per litter (eq/L) * of major ions that are typically found in freshwater (e.g., rivers and lakes) [16] and seawater [17] worldwide.

Freshwater		**Seawater**	
Cations	**Anions**	**Cations**	**Anions**
Ca^{2+} (52.6%)	HCO_3^- (67.1%)	Na^+ (76.34%)	Cl^- (90.12%)
Mg^{2+} (24.0%)	SO_4^{2-} (16.3%)	Mg^{2+} (18.57%)	SO_4^{2-} (9.45%)
Na^+ (19.3%)	Cl^- (15.4%)	Ca^{2+} (3.42%)	HCO_3^- (0.28%)
K^+ (4.1%)		K^+ (1.67%)	Br^- (0.14%)

*The values could vary depending on the specific geochemistry of the sites. Note that freshwater is defined as having the salinity of less than 1000 mg/L (approx. 0.017 M as NaCl). Salinity up to 3000 mg/L (approx. 0.051 M as NaCl) is fresh to brackish water, up to 5000 mg/L (approx. 0.086 M as NaCl) is brackish water, up to 35,000 mg/L (approx. 0.6 M as NaCl) is saline water, and over 35,000 mg/L is hyper-saline water. Typical sweater has a salinity of about 35,000 mg/L [18].

Table S3. Thermodynamic parameters of BSA/PFHpA binding obtained from binding Van 't Hoff plot (Fig. S6).

Added salt	ΔH (kJ/mol)	ΔS (kJ/mol/K)	R^2	T (K)	ΔG (kJ/mol)	TΔS (kJ/mol)
no salt	15.6	0.138	0.98	293	-24.8	40.3
				300	-25.7	41.3
				310	-27.1	42.7
				318	-28.2	43.8
NaCl (150 mM)	21.1	0.143	0.94	293	-22.8	43.9
				300	-23.8	44.9
				310	-25.3	46.4
				318	-26.5	47.6
KCl (150 mM)	16.1	0.133	0.99	293	-22.9	39.0
				300	-23.9	39.9
				310	-25.7	41.3
				318	-26.3	42.3
$CaCl_2$ (50 mM)	16.9	0.140	0.94	293	-24.2	41.1
				300	-25.2	42.1
				310	-26.6	43.5
				318	-27.7	44.6
$MgCl_2$ (50 mM)	17.5	0.143	0.99	293	-24.3	41.8
				300	-25.3	42.8
				310	-26.7	44.3
				318	-27.9	45.4
Na_2SO_4 (60 mM)	17.9	0.139	0.99	293	-23.9	41.8
				300	-24.9	42.8
				310	-26.3	44.3
				318	-27.5	45.4
Na_2SO_4 (75 mM)	17.8	0.138	0.99	293	-24.1	41.8
				300	-25.1	42.8
				310	-26.5	44.3
				318	-27.6	45.4

Table S4. Model derived (K_{as_app}) and experimentally obtained (K_{a_Hill}) binding constants of PFHpA/BSA in 20 mM MOPs or 20 mM sodium phosphate buffer (pH 7, 20°C) in the absence and in the presence of additional salts (i.e., NaCl, KCl, $CaCl_2$, $MgCl_2$, and Na_2SO_4).

Background buffer	Additional salt	$K_{a_Hill} \times 10^{-4}$ (L/mol)	$K_{as_app} \times 10^{-4}$ (L/mol)
20 mM MOPs	-	2.67 ± 0.17	2.60
20 mM MOPs	150 mM NaCl	1.08 ± 0.04	1.36
20 mM MOPs	300 mM NaCl	0.98 ± 0.04	9.19
20 mM MOPs	150 mM KCl	1.21 ± 0.05	1.36
20 mM MOPs	5 mM $CaCl_2$	2.81 ± 0.28	3.16
20 mM MOPs	50 mM $CaCl_2$	2.24 ± 0.13	2.09
20 mM MOPs	100 mM $CaCl_2$	1.76 ± 0.09	1.51
20 mM MOPs	5 mM $MgCl_2$	3.33 ± 0.28	3.16
20 mM MOPs	50 mM $MgCl_2$	2.12 ± 0.10	2.09
20 mM MOPs	100 mM $MgCl_2$	1.56 ± 0.06	1.51
20 mM MOPs	60 mM Na_2SO_4	1.10 ± 0.10	1.22
20 mM MOPs	75 mM Na_2SO_4	1.04 ± 0.09	1.08
20 mM $NaH_2PO_4-Na_2HPO_4$	-	2.43 ± 0.19	2.13
20 mM $NaH_2PO_4-Na_2HPO_4$	10 mM NaCl	1.96 ± 0.14	2.03
20 mM $NaH_2PO_4-Na_2HPO_4$	30 mM NaCl	1.92 ± 0.18	1.85
20 mM $NaH_2PO_4-Na_2HPO_4$	80 mM NaCl	1.45 ± 0.10	1.52

Application of binding model

Fluorescence profiles of PFAS/BSA were measured at range of ionic strength with different salt species to quantify the binding constants (K_{a_Hill}) based on the Hill equation. Model parameters (i.e., K_{as1}, K_{as2}, K_{A1}, and K_{A2}) wewre fitted assuming that K_{as_app} is equivalent to K_{a_Hill}. To do this the sum of the squared difference K_{a_Hill} and K_{as_app} was minimized. Parameter fits for an individual salt condition are likely not unique. As such, model parameters are determined by simultaneously fitting multiple salt conditions, including monovalent and divalent salt conditions. Parameters uncertainties were derived from the covariance matrix of the fit, assuming normally distributed residuals. Notably, since the K_{as_app} value is calculated from the fitted parameters rather than being directly fitted, its associated uncertainty is not presented. Once the model parameters (i.e., K_{as1}, K_{as2}, K_{A1}, and K_{A2}) for a given PFAS are determined, they can be used for prediction of binding affinities at a range of salt conditions.

Based upon the model predicted binding affinity it is possible to quantitatively estimate concentrations (and percentage) of PFAS bound to proteins under environmental or physiological salt conditions.

$$[PS] + [P_{app}] = [P_{total}] \tag{12}$$

$$[PS] + [S] = [S_{total}] \tag{13}$$

as shown before, [PS] is the concentration of PFAS/protein complex, $[P_{app}]$ is the concentration of unbound protein as shown in Eq. (10), and [S] is the concentration of unbound PFAS. $[P_{total}]$ is the concentration of total protein, $[S_{total}]$ is the concentration of total PFAS. Whenever the $[P_{total}]$ and $[S_{total}]$ for a specific system is known and combining with K_{as_app}, the concentration of PFAS/protein complex [PS] can be derived. The percentage of bound PFAS is given by:

$$\%\ of\ bound\ PFAS = \frac{[PS]}{[S_{total}]}$$

$$= \frac{K_{as_app}P_{total} + K_{as_app}S_{total} + 1 - \sqrt{(K_{as_app}P_{total} + K_{as_app}S_{total} + 1)^2 - 4K_{as_app}{}^2 P_{total}S_{total}}}{2K_{as_app}S_{total}} \tag{14}$$

Note that the expression of [PS] in Eq. 14 is based on rearranging Eq. 11, 12, and 13, which shown as: $K_{as_app}\,[PS]^2 - \left(K_{as_app}P_{total} + K_{as_app}S_{total} + 1\right)[PS] + K_{as_app}P_{total}S_{total} = 0$. Based up the quadratic formular, the roots for quadratic equation (i.e., $ax^2 + bx + c = 0$) are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Given that the [PS] should be no larger than $[S_{total}]$, [PS] is expressed in term of $\frac{K_{as_app}P_{total} + K_{as_app}S_{total} + 1 - \sqrt{(K_{as_app}P_{total} + K_{as_app}S_{total} + 1)^2 - 4K_{as_app}{}^2 P_{total}S_{total}}}{2K_{as_app}}$.

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